Conditionals & Repetitions – Python

Extra Exercises

WI1102 Computational Thinking Teaching Team

Institut Teknologi Bandung © 2026





Conditionals

Exercises

- For the following exercises, practice creating:
  - **Flowchart** or **Pseudocode** (please choose, or determined by the class lecturer)
  - Corresponding **Python** program

Exercise 1. Largest of Two Numbers

- Write a program that reads two integers, for example A and B, and writes down which of the two numbers is the largest.
- Cases:
  - If A > B, then the largest number = A
  - If A < B, then the largest number = B
  - If A = B, then the largest number is A or B (meaning the output will be the same as one of the 2 cases above)
- Are these 3 cases or only 2 cases?

Exercise 2. Form of Water

- Write a program that receives the temperature of water (in degrees Celsius) and prints the form of the water to the screen, whether it is “frozen”, “liquid”, or “steam”.
  - If the water temperature is <= 0 degrees, then print “frozen”
  - If the water temperature is > 0 and < 100 degrees, then print "liquid"
  - If the water temperature is >= 100, then print "steam"

Exercise 3. Total of Serial Resistance

- Write a program that accepts 3 resistances (R1, R2, R3) and produces a total resistance (RT) if connected in series.

$$\mathbf{RT = R1 + R2 + R3}$$

- R1, R2, and R3 cannot be negative. If just one resistance is negative, then the total resistance cannot be calculated and the error message "Total resistance cannot be calculated" will appear on the screen.

Exercise 4. Rank of 3 Numbers

- Given 3 integers that are read from the keyboard, for example A, B, C. Assume that these three numbers are **different**.
- Print on the screen the three numbers in order from largest to smallest.
- Example: A = 1, B = -1, C = 2

  Then it displays on the screen: 2 1 -1

Exercise Alternative Solutions

For instructors only

Exercise 1. Largest of Two Numbers

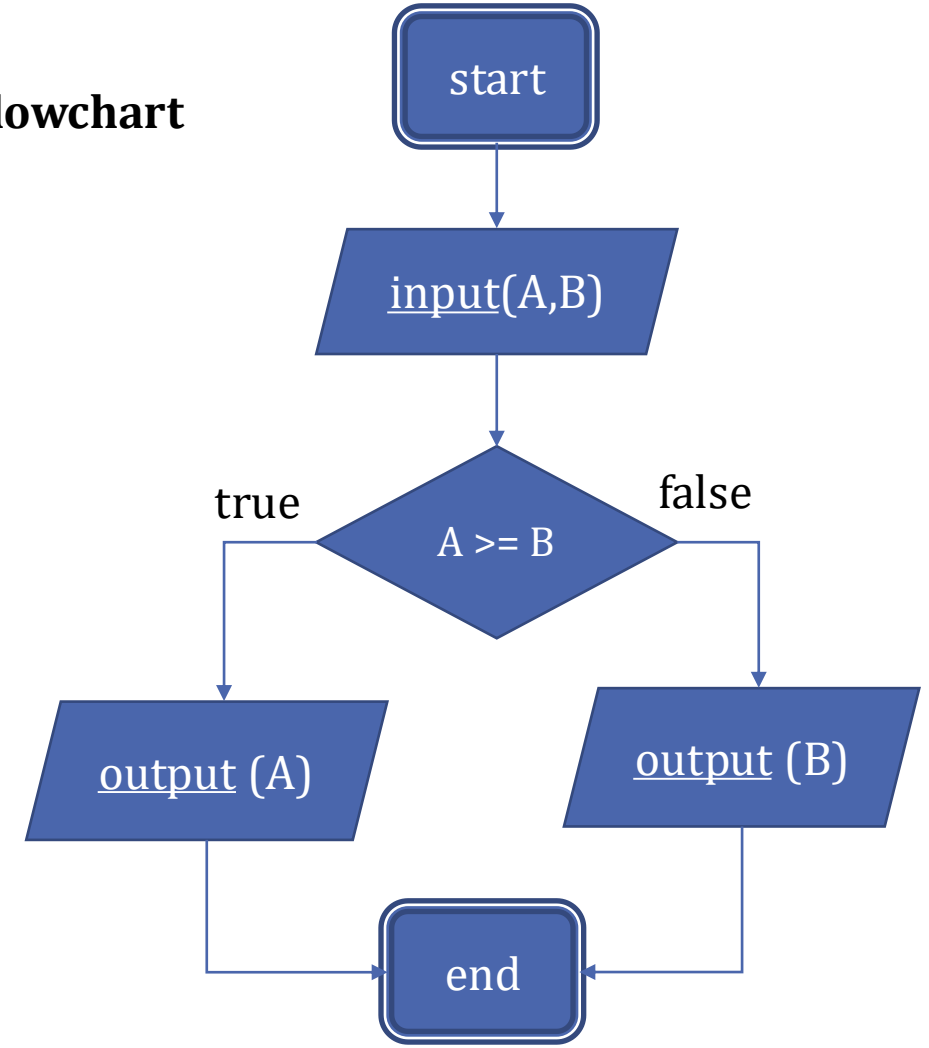
- There are only 2 cases, the alternatives:
  - If A >= B, then the largest number is B
  - If A < B (complement of the case above), then the largest number = B

Exercise 1. Pseudocode + Flowchart

Pseudocode

```
input(A, B)
if (A >= B) then
        output(A)
else { A < B }
        output(B)
```

Flowchart



Exercise 1:. Python

```
# Program Max2Integer
# Print the largest value of the input 2 integers

# DECLARATIONS
# A, B : int

# ALGORITHM
A = int(input())
B = int(input())

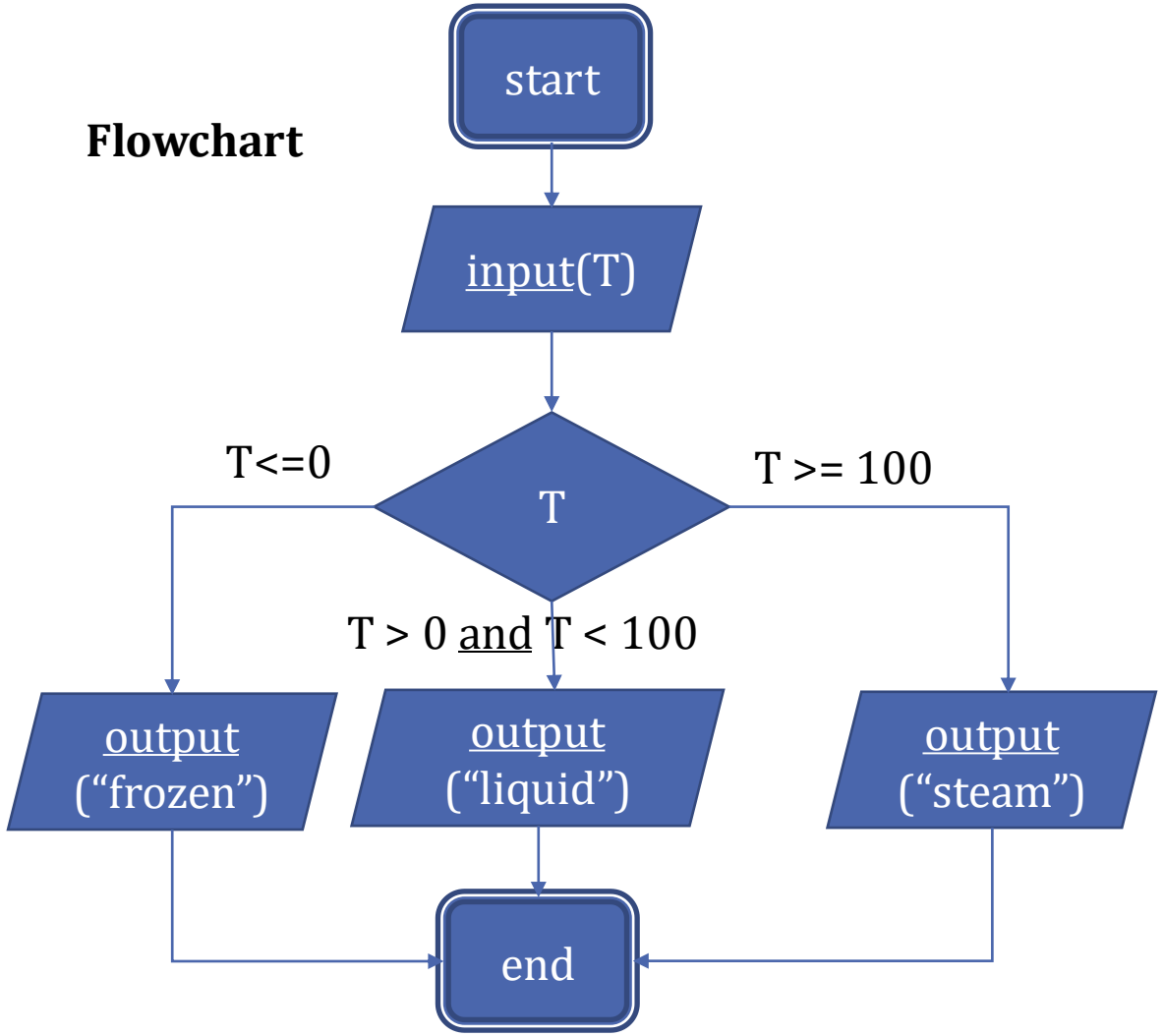
if (A >= B):
    print(A)
else: # A < B
    print(B)
```

Exercise 2. Pseudocode + Flowchart

Pseudocode

```
input(T)   { T is water temperature }
if (T <= 0) then
      output("frozen")
else if (T > 0 and T < 100) then
      output("liquid")
else { T >= 100 }
      output("steam")
```

Flowchart



Exercise 2. Python

```
# Program WaterForm
# Print the form of water depending on the temperature input

# DECLARATIONS
# T : float

# ALGORITHM
T = float(input())

if (T <= 0):
    print ("frozen")
elif (T > 0 and T < 100):
    print ("liquid")
else: # T >= 100
    print ("steam")
```

Exercise 3. Total of Serial Resistance

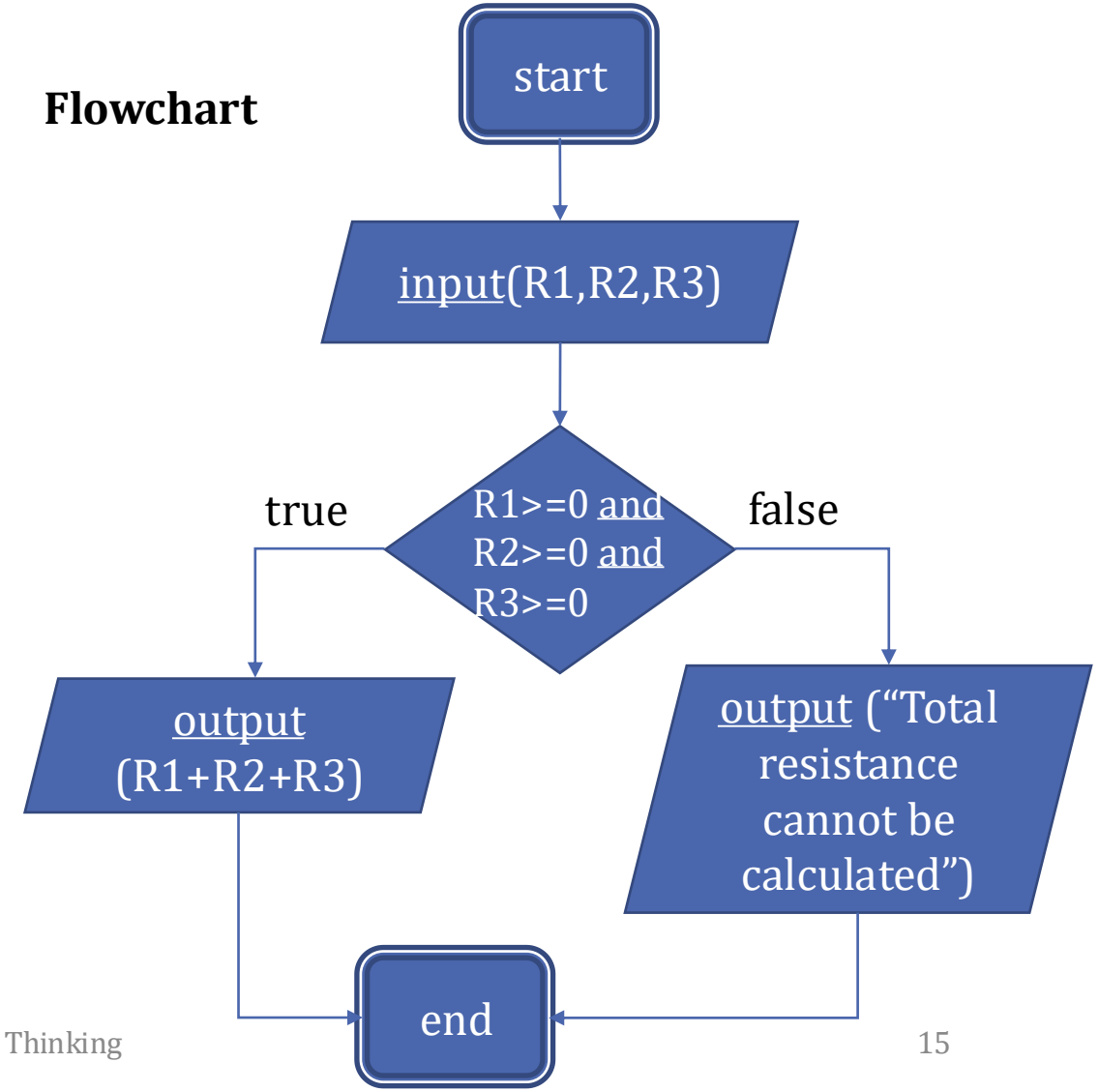
- Cases:
  - If R1 >= 0 and R2 >= 0 and R3 >= 0, then calculate the total resistance using the formula: RT = R1 + R2 + R3
  - If not, namely R1 <= 0 or R2 < 0 or R3 < 0, then write the error message "Total resistance cannot be calculated"

Exercise 3. Pseudocode + Flowchart

Pseudocode

```
input (R1, R2, R3)
if (R1 >= 0) and (R2 >= 0) and (R3 >= 0) then
        output(R1+R2+R3)
else {R1 < 0 or R2 < 0 or  R3 >= 0}
        output("Total resistance cannot be calculated")
```

Flowchart



Exercise-3: Python

```
# Program SerialResistance
# Calculate total of serial resistance based on resistance R1, R2, R3

# DECLARATIONS
# R1, R2, R3 : float

# ALGORITHM
R1 = float(input()); R2 = float(input()); R3 = float(input())

if ((R1 >= 0) and (R2 >= 0) and (R3 >= 0)):
    print (R1 + R2 + R3)
else: # R1 < 0 atau R2 < 0 atau R3 < 0
    print("Total resistance cannot be calculated")
```

Exercise 4. Rank of 3 Numbers (1)

Several alternative of solutions:

1. Enumeration of relationships between A, B, C there are 6 cases:
    1. A > B > C → A B C
    2. A > C > B → A C B
    3. … (continue)
2. Determine the maximum value between 2 numbers, then determine the position relative to the 3rd number. Example:
    1. A > B, then A must be to the "left" of B. Just determine the position of C.
        1. If B > C, then: A B C
        2. If B < C, then C must be to the "left" of B, depending on C's position relative to A:
            1. If A > C, then: A C B
            2. If A < C, then: C A B
    2. A < B … (continue)

Exercise-4: Rangking 3 bilangan (2)

Several alternative of solutions:

3. Find the largest value of A, B, C, then determine the position of the other 2 numbers
    1. A > B and A > C, then:
        1. If B > C, then: A B C
        2. If B < C, then: A C B
    2. B > A and B > C, maka:
        1. If A > C, then: B A C
        2. If A < C, then: B C A
    3. … (continue)
4. … other alternatives??

Exercise-4: Python

Alternative Solution 1

```
# Program Rank3Numbers
# Accepts input of 3 numbers: A, B, C
# Print all three numbers in descending order
# Assumptions A, B, C are different

# DECLARATIONS
# A, B, C : int

# ALGORITHM
A = int(input()); B = int(input()); C = int(input())

if (A > B) and (B > C):
    print (A, " ", B, " ", C)
elif (A > C) and (C > B):
    print (A, " ", C, " ", B)
elif (B > A) and (A > C):
    print (B, " ", A, " ", C)
elif (B > C) and (C > A):
    print (B, " ", C, " ", A)
elif (C > A) and (A > B):
    print (C, " ", A, " ", B)
else: # C > B > A
    print (C, " ", B, " ", A)
```

```
# Alternative for expression:
if (A > B > C):
    print (A, " ", B, " ", C)
elif (A > C > B):
    print (A, " ", C, " ", B)
elif (B > A > C):
    print (B, " ", A, " ", C)
elif (B > C > A):
    print (B, " ", C, " ", A)
elif (C > A > B):
    print (C, " ", A, " ", B)
else: # C > B > A
    print (C, " ", B, " ", A)
```

Repetition/Looping

Exercise

- For all of the following exercises:
  - Practice creating **Python** programs with the best or required repetition notation
  - Also make a **flowchart/pseudocode** (depending on what is requested by the class lecturer)

Exercise 1

- Create an program that reads a positive integer value, for example N, and adds (and displays) all multiples of 5 between 1 and 1. N.
- Example:

No	Input N	Output	Note
1	5	5	There is only 1 multiple of 5 between 1 and 5, namely 5
2	26	75	Multiples of 5 between 1 and 26 are 5, 10, 15, 20, 25 5+10+15+20+25 = 75
3	4	0	There are no multiples of 5 between 1 and 14

Exercise 2

- Write a program that reads how many students are in the class, for example N (Assumption: N > 0, no need to check)
- Next, read the N characters that represent the grades of the WI1102 assignment. Possible assignment grades are: ‘A’, ‘B’, ‘C’, ‘D’, ‘E’, ‘F’. Assume the input value is always true.
- If a student gets a grade: A', 'B', 'C', or 'D', then the student is declared to have passed; whereas if you get an 'E' or 'F' then the student is declared not to have passed.
- Write on the screen how many students passed and how many did not pass.

Exercise 2. Examples

No	Input N	Input of WI1102 grades	Output
1	7	A B C A A E D	Pass = 6 Not pass = 1
2	5	A B B A A	Pass = 5 Not pass = 0
3	5	E E E E E	Pass = 0 Not pass = 5

Exercise 3

- Write a program to read a set of positive integers. Data reading is terminated if the user enters a negative value.
- Next, print out how many even and odd numbers there are.
- 0 is an even number.

No	Input	Output	Note
1	7 8 0 9 10 -1	Even = 3 Odd = 2	
2	4 10 6 -111	Even = 3 Odd = 0	
3	5 17 -234	Even = 0 Odd = 2	
4	-99	Even = 0 Odd = 0	No positive numbers are entered
All negative number input ends data reading			

Exercise 4. Lagu Anak Ayam

- Do you still remember the song **Anak Ayam**??

Anak ayam turunlah 5
Mati satu tinggallah 4
Mati satu tinggallah 3
Mati satu tinggallah 2
Mati satu tinggallah 1
Mati satu tinggal induknya

Anak ayam turunlah 1
Mati satu tinggal induknya

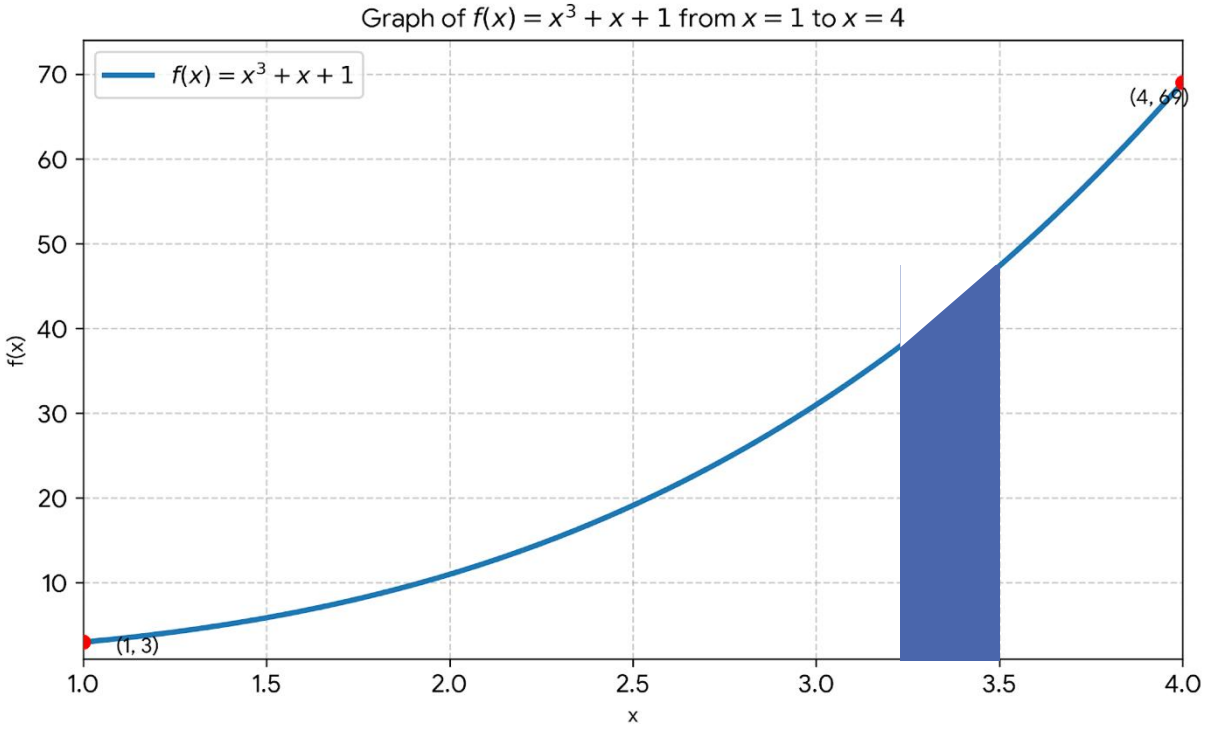


Anak ayam turunlah **N**
Mati satu tinggallah **N-1**
Mati satu tinggallah **N-2**
….
Mati satu tinggallah **1**
Mati satu tinggal induknya

- Create a program that accept as input a positive integer, for example N (assuming N>0), and write the lyrics to the Anak Ayam song above

Exercise 5. Calculating the Area under the curve

- To calculate the area of a curve formed by a formula, it can be done using an integral using a numerical approach.
- The numerical approach will cut the area at certain intervals, then calculate the area of each piece of the area using the trapezoid formula repeatedly.
- Write a program for the following problem.



To calculate the area under the curve defined by the function $f(x) = x^3 + x + 1$ from $x = 1$ to $x = 4$, we can divide the range into intervals (e.g., 0.25). The smaller the interval, the more detailed the result.
The total area is obtained by calculating the area of all the trapezoids formed by these intervals.

Task:
Write a program to calculate the area under the curve defined by the function $f(x) = x^3 + x + 1$ from $x = a$ to $x = b$ using a specified interval (delta), where a, b, and delta are user inputs.

Assumptions:
$a < b$; $a \geq 0$; $b > 0$; $delta > 0$

Exercise 6

- The BMKG (Badan Meteorologi, Klimatologi, dan Geofisika, Meteorology, Climatology, and Geophysics Agency) of Bandung City requires a program to calculate several basic statistics regarding air temperature in Bandung City over the course of a month, specifically:
  - Average temperature
  - Highest temperature
  - Lowest temperature
- Create a program that accepts the number of days in the month—denoted as N (where N can be 28, 29, 30, or 31; assume the input is valid)—then accepts the air temperatures from day 1 through day N of that month and outputs the aforementioned statistics.

Exercise Alternative Solutions

For instructors only
Complete alternative solutions (Python, pseudocode, and flowchart) are provided only for Exercise 1.
Alternative solutions in Python only are provided for Exercises 2 through 4.

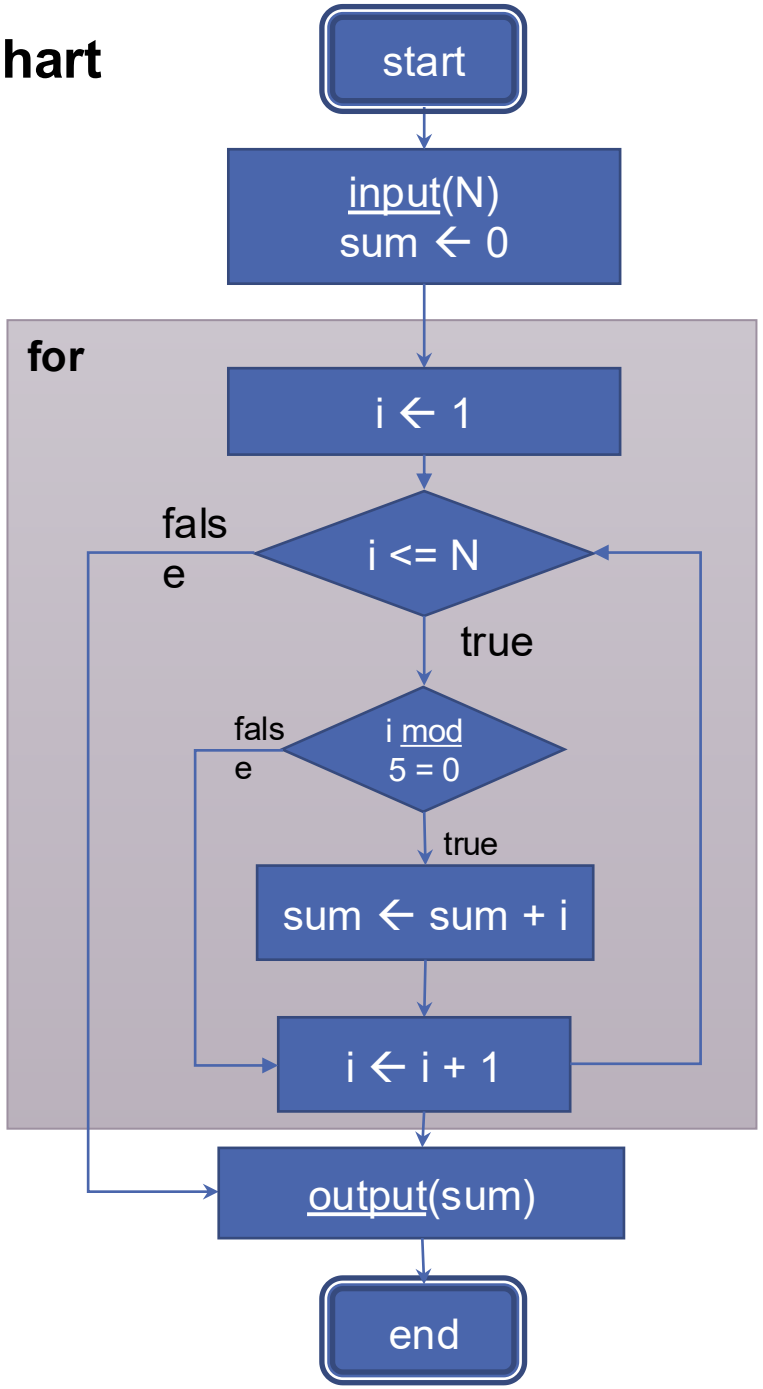
Exercise 1 Pseudocode+Flowchart

The most appropriate loop is the **for** loop, because the range of values for the iteration is known precisely—namely, from 1 to N.

pseudocode

```
input(N)  { Initialization }
sum ← 0  { Initialization }
i traversal [1..N]
    { aksi }
    if (i mod 5 = 0) then {cek kelipatan 5}
        sum ← sum + i
output(sum)  { terminasi }
```

flowchart



Exercise 1 Python

```
# Program SumKelipatan5
# Menerima masukan N bilangan integer positif
# dan menampilkan penjumlahan bilangan kelipatan 5 dari 1 s.d. N

# DECLARATIONS
# N, i, sum : int

# ALGORITHM
# Masukan grade N
N = int(input())

# Menghitung penjumlahan bilangan kelipatan 5 antara 1 s.d. N
sum = 0
for i in range(1,N+1):
    if (i % 5 == 0):
        sum = sum + i

# Terminasi
print(sum)
```

Exercise 2. Discussion

- The most appropriate loop is the **for** loop, as the range of values for the iteration is known precisely—specifically, from 1 to N.
- Using a **while** loop is less suitable because there is no possibility of an "empty" case.

Exercise 2 Python

```
# Program Tugas
# Accepts N assignment grades (A through F) as input
# and outputs the number of passing and not passing (failing) grades.

# DECLARATIONS
# N, i : int
# grade : char
# pass, notpass : int

# ALGORITHM
# Entry N
N = int(input("Number of students = "))

# Enter the scores and simultaneously calculate pass/not pass status.
pass = 0; notpass = 0
print("Enter each student's grade..")
for i in range(1,N+1):
    grade = input("Student-" + str(i) + " = ")
    if (grade == 'A' or grade == 'B' or grade == 'C' or grade == 'D'):
        pass = pass + 1
    else: # grade == 'E' atau 'F'
        notpass = notpass + 1

# Termination
print("Pass = " + str(pass))
print("Not pass = " + str(notpass))
```

Exercise 3. Discussion

- The most appropriate loop to use is the **while** loop.
  - There is a possibility that the action is never performed at all (an empty case)—specifically, if the initial value entered by the user is negative.
- The **for** loop is unsuitable because the range of values is undefined.

Exercise 3 Python

```
# Program EvenOdd
# Accept a series of integers until the user enters a negative value,
# then display the count of even and odd numbers.


# DECLARATIONS
# X, countEven, countOdd : int

# ALGORITHM
countEven = 0                       # Initialization
countOdd = 0                        # Initialization

X = int(input())                    # First-Elmt
while (X >= 0):
    # cek Odd atau Even
    if (X % 2 == 0): # X is even
        countEven = countEven + 1
    else: # X is odd
        countOdd = countOdd + 1
    X = int(input())                # Next-Elmt

# Terminasi
print("Number of even numbers = " + str(countEven))
print("Number of odd numbers = " + str(countOdd))
```

Exercise 4. Discussion

- A loop is used to print the phrase “Mati satu tinggallah..."
- The loop prints the number of chicks, counting down from N-1 to 1.
- If N = 1, the phrase “Mati satu tinggallah..." is never printed → an empty case.
  - **for**: the empty case must be prevented from entering the loop by using an if-else statement.
  - **while**: the safest option, as the empty case is handled by specifying the loop condition at the start.

Exercise 4: Python for

```
# Program AnakAyam-1
# Write Anak Ayam song based on input number of
# chicks
# Version using for

# DECLARATIONS
# N, i : int

# ALGORITHM
N = int(input("Enter the number of chicks = "))

print("Anak ayam turunlah " + str(N))
if (N > 1):
    for i in range(N-1,0,-1):
        print("Mati satu tinggallah " + str(i))

print("Mati satu tinggal induknya")
```

Exercise 4: Python while

```
# Program AnakAyam-2
# Write Anak Ayam song based on input number of
# chicks
# Version using while

# DECLARATIONS
# N, i : int

# ALGORITHM
N = int(input("Enter the number of chicks = "))

print("Anak ayam turunlah " + str(N))
i = N-1
while (i >= 1):
    print("Mati satu tinggallah " + str(i))
    i-=1  # i=i-1
# i < 1

print("Mati satu tinggal induknya")
```